PROJECT	DOCUMENT	REVISION	DATE
APOLLO TRACK POD	FABRICATION BLUEPRINT	002C	2026-07-22

UNITS	SHEETS
MM [IN]	7 + SCHEDULES + GUIDES

MODIFICATION TYPE — ARTICULATED SPLIT-FRAME

ARTICULATED "SPLIT-FRAME" SUSPENSION MOD

Converts a rigid rubber-track pod on a hub-motor scooter (motor inside the Ø180 10T sprocket on a static Ø10 axle between fork legs, two bearing-mounted lower idlers, lugged rubber band) into an independently articulating suspension: a central pivot below the drive hub, a leading and a trailing arm, and two coil-over shocks. Rev 002c: every load-bearing dimension is measured and baked in as a number — the only open measurement left is the guide-lug height.

§0 READ THIS BEFORE ANYTHING ELSE

The pod you bought (photos: sprocket + two idlers + rubber band, sold as a wheel-replacement track for mowers / small ATVs / mobility platforms) has **no external steel frame**. The idlers hang off an internal molded/plate structure hidden behind the sprocket, and dimensions vary between sellers even when the listings look identical. That structure is exactly what this mod removes and replaces, so:

- **This drawing set is parametric.** Datums A–J on Sheet 0 are measured off *your* pod with calipers and a ruler. Every dimension on Sheets 1–6 is a formula of those datums. Fill the table in, then transfer numbers to the drawings.
- **Disassemble one pod completely first.** Photograph the internal frame, the idler axle hardware, and — critically — however the pod stops itself from spinning around the drive axle (anti-rotation). Your carrier plates must reproduce that interface.
- **Build one pod, test it, then replicate.** Don't cut four sets of parts before the first pod has survived the Sheet-7 test protocol.

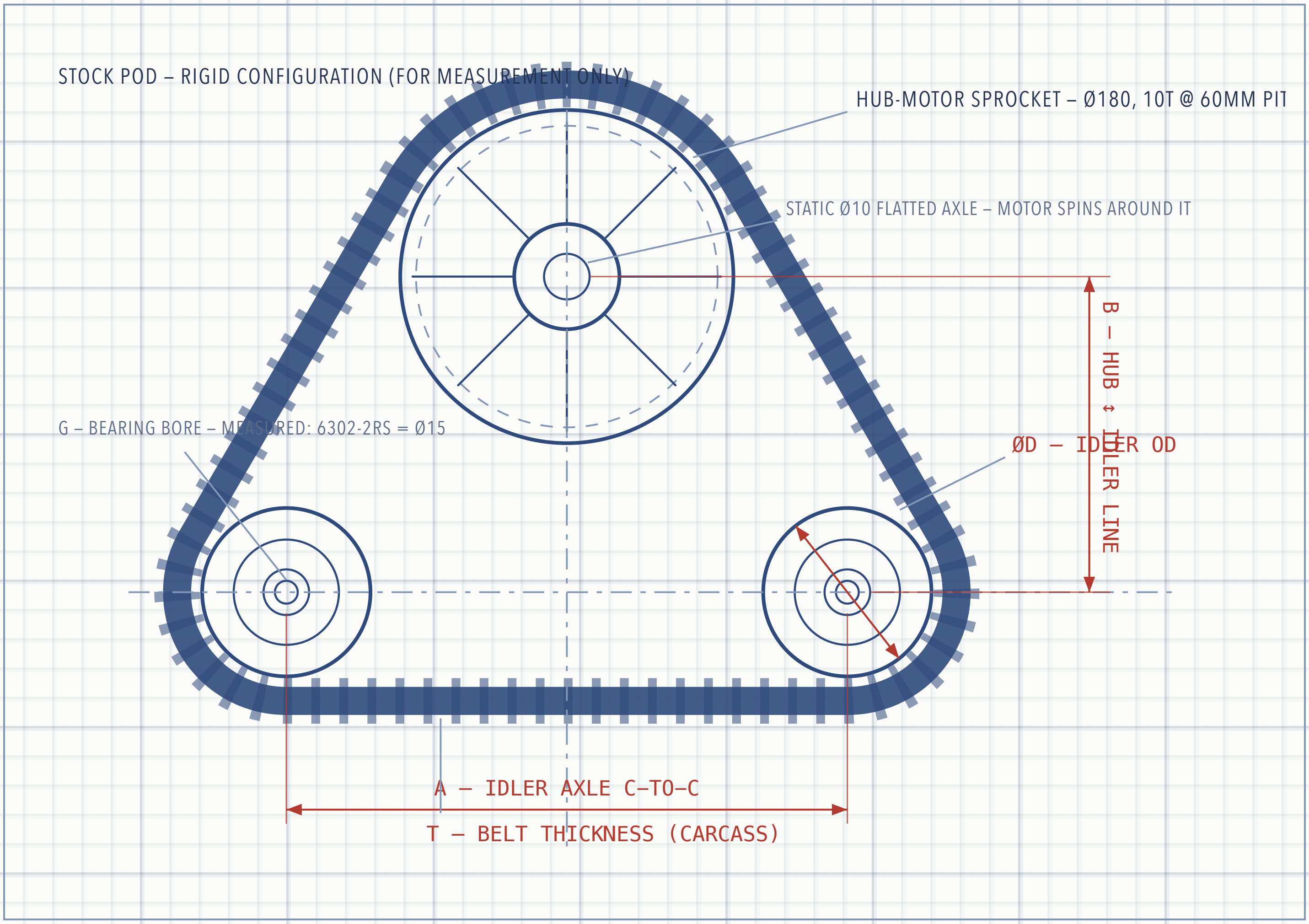
DESIGN RISK – READ

Track retention is the failure mode of this mod. A rigid triangular pod keeps the belt at constant wrap length. Once the arms articulate, the belt path length changes a few millimetres through travel and the belt *will* try to walk off the idlers under side load. The design compensates three ways, and all three are mandatory: travel limited to ±15°, spring preload keeping the belt tensioned through droop, and the trailing-arm tensioner (Sheet 6) set correctly. Deep-lug tracks like yours help; they are not sufficient alone.

Anti-rotation — resolved in Rev 002. The carriers bolt directly to the scooter-fork legs, and their axle slots grip the hub-motor's flatted Ø10 axle exactly like a scooter torque arm. Drive torque reacts straight into the fork; no separate link needed. The old watch-item here is retired: Rev 003 re-measured **both fork gaps at 140 mm** (the earlier 120 front reading was wrong), so the 118 mm belt has 11 mm per side front and rear. Still verify the belt tracks true at first assembly.

SHEET 0 DATUM MEASUREMENTS – TAKE THESE OFF YOUR POD

Set the pod on a flat floor, belt tensioned as delivered. Measure with calipers where possible. Record everything in the table; the formula column on later sheets consumes these letters.



SHEET 0	SCALE	REV	UNITS
DATUM SURVEY – STOCK POD	NTS	002	MM

DATUM TABLE – FILL IN WITH CALIPERS

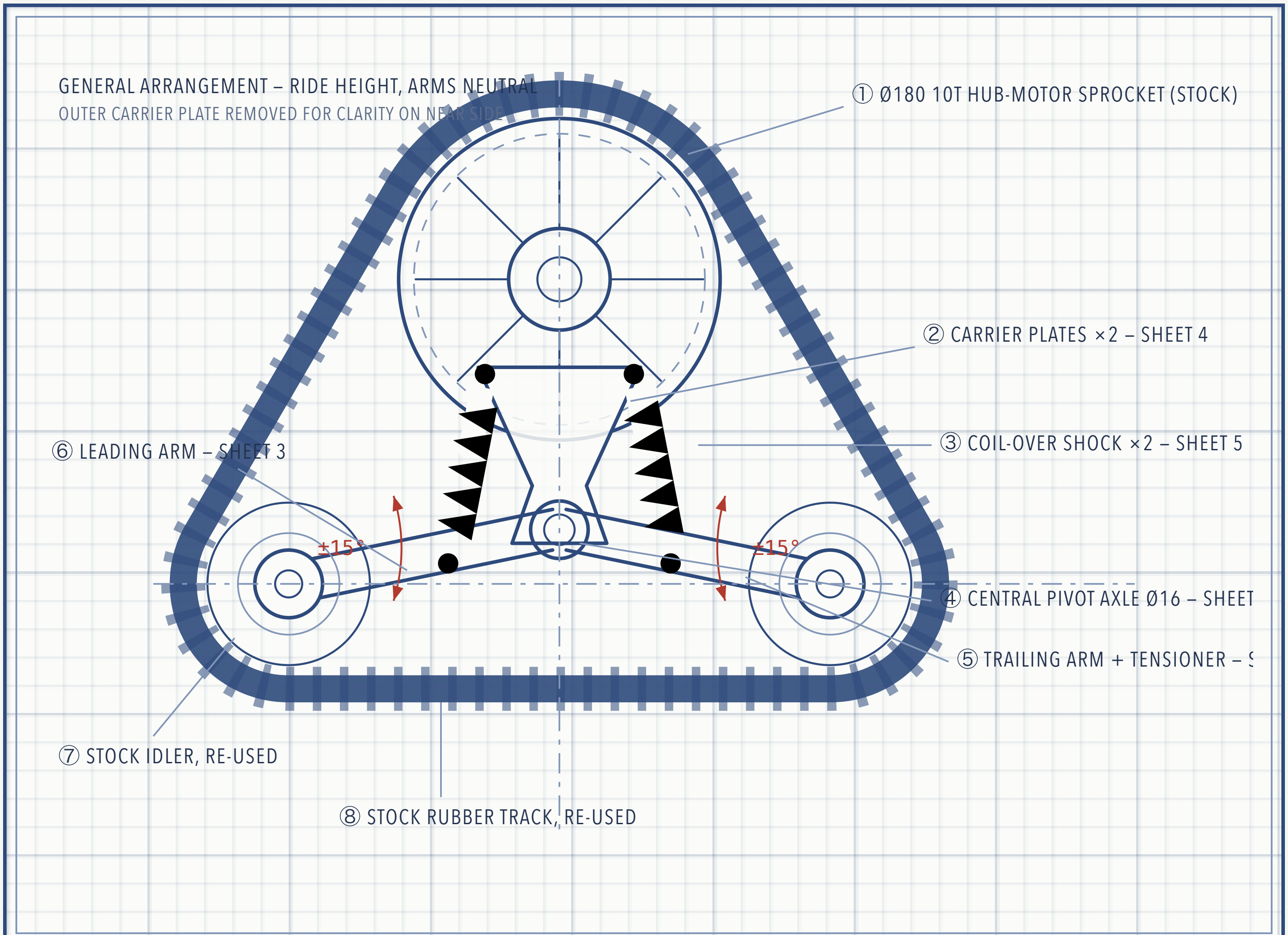
DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
L	Belt cord-line circumference × overall width	Links × pitch off the belt spec	—	**1080 × 118** (60 mm pitch × 18 links, 2.1 kg)
A	Idler axle centre-to-centre distance	Solved from L along the belt *cord line* (10T × 60 = cord Ø191, 5.5 above the Ø180 face) — tensioner absorbs the residue	230–300	**222.2** (solved)

DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
B	Drive-hub centre height above idler axle line	Square + ruler from floor; B = hub height − idler height	150–200	**170**
D	Idler wheel outside diameter	Calipers across the wheel	90–120	**108** (WJ wheel)
G	Idler bearing bore (axle Ø)	Read bearing number off seal	15–20	**15** — 6302-2RS (15×42×13); caliper read 14.86 = Ø15 nominal
fork	Fork-leg inner gap, front / rear pod	Inner face to inner face	—	**140 / 140** — leg thickness **4** (gaps re-measured 2026-07-20, Rev 003 — was 120/140)
axle	Hub-motor axle	—	—	**Ø10, flatted, static** — motor spins around it
F	Belt inner width between guide lugs	Derived — wheel and belt are a matched set	60–90	**62** (H + 4); lug height ___ (20 assumed, owner est. 10–20)
H	Idler hub width (bearing-to-bearing over the wheel)	Calipers over the wheel hub	40–60	**58** (57.85)
T	Belt carcass thickness (not incl. lugs)	Calipers on a straight span edge	8–14	**~12** (confirmed)

*Typical ranges for this class of pod — for sanity-checking only, never for cutting. Derived values used on later sheets (exact, echoed by the .scad on every render): A = 222.2 · P = B − 38 = 132 · C = $\sqrt{(A/2)^2 + 38^2}$ = 117.4 · neutral arm droop 18.9° · shock station a = 0.46·C = 54.0 · upper shock eye at (±53, −9) from hub centre · keel centre 100 below hub centre (Ø12×1.5 tube) · wheel travel +30 / −27.5. The 38 mm pivot lift (was 20, then 30) keeps everything crossing the belt zone clear of the bottom run at full bump; 38 is near the hard limit of 42 where the keel window closes.

SHEET 1 GENERAL ARRANGEMENT – HOW THE MECHANISM WORKS

The rigid internal frame is replaced by two identical steel **carrier plates** that hang from the scooter-fork legs: each plate slots over the hub-motor's static Ø10 flatted axle (doubling as a torque arm) and takes one M8 bolt into the leg 52 mm above (Rev 002 — no carrier bearings, no anti-rotation link). A **Ø16 pivot axle** spans the plates 132 mm below the hub. The **leading arm** (front idler) and **trailing arm** (rear idler) are fork-shaped weldments that swing on that axle like scissor halves. Each arm carries a **coil-over shock** reacting against tabs on the carrier inner faces. Wheel hits a bump → arm swings up → shock compresses. Each side is fully independent.



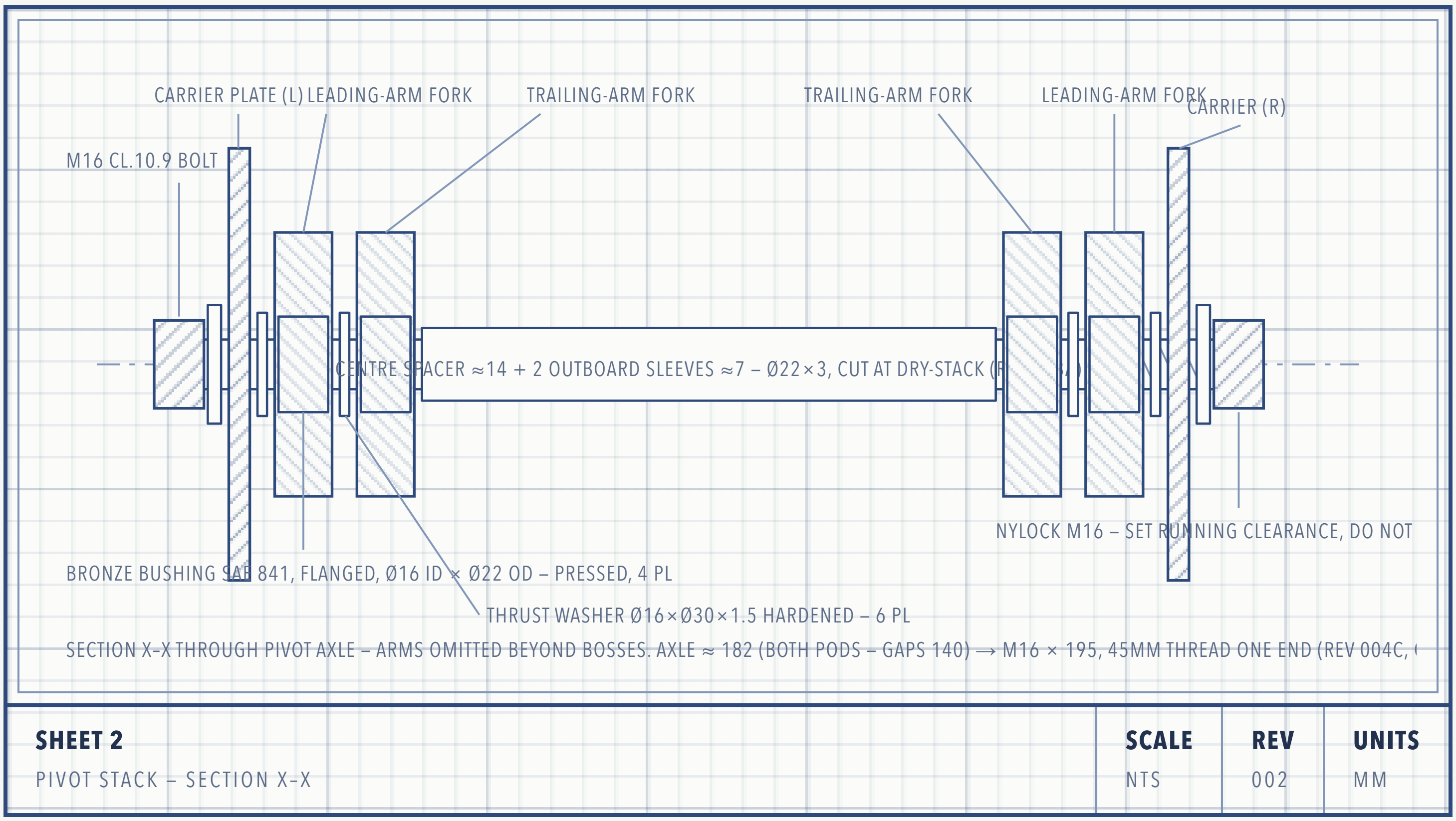
SHEET 1	SCALE	REV	UNITS
GENERAL ARRANGEMENT	NTS	002	MM

Arms drawn at neutral. Each wheel gets **+30 mm bump / –27.5 mm droop** at ±15° (C = 117.4) — a huge improvement over the rigid pod's zero. At full bump the belt's bottom run rises ~30 mm toward the pivot; everything that crosses the belt zone clears it (pivot spacer 29 mm, keel 62 mm above the assumed lug tops), and the carrier plates themselves ride at |z| = 74 (gaps 140, legs 4 mm — Rev 003)

— outside the belt's 59 mm half-width, so they *cannot* touch the track at any articulation. Verify with the OpenSCAD clearance guard (all lines must read PASS).

SHEET 2 CENTRAL PIVOT AXLE – STACK-UP SECTION

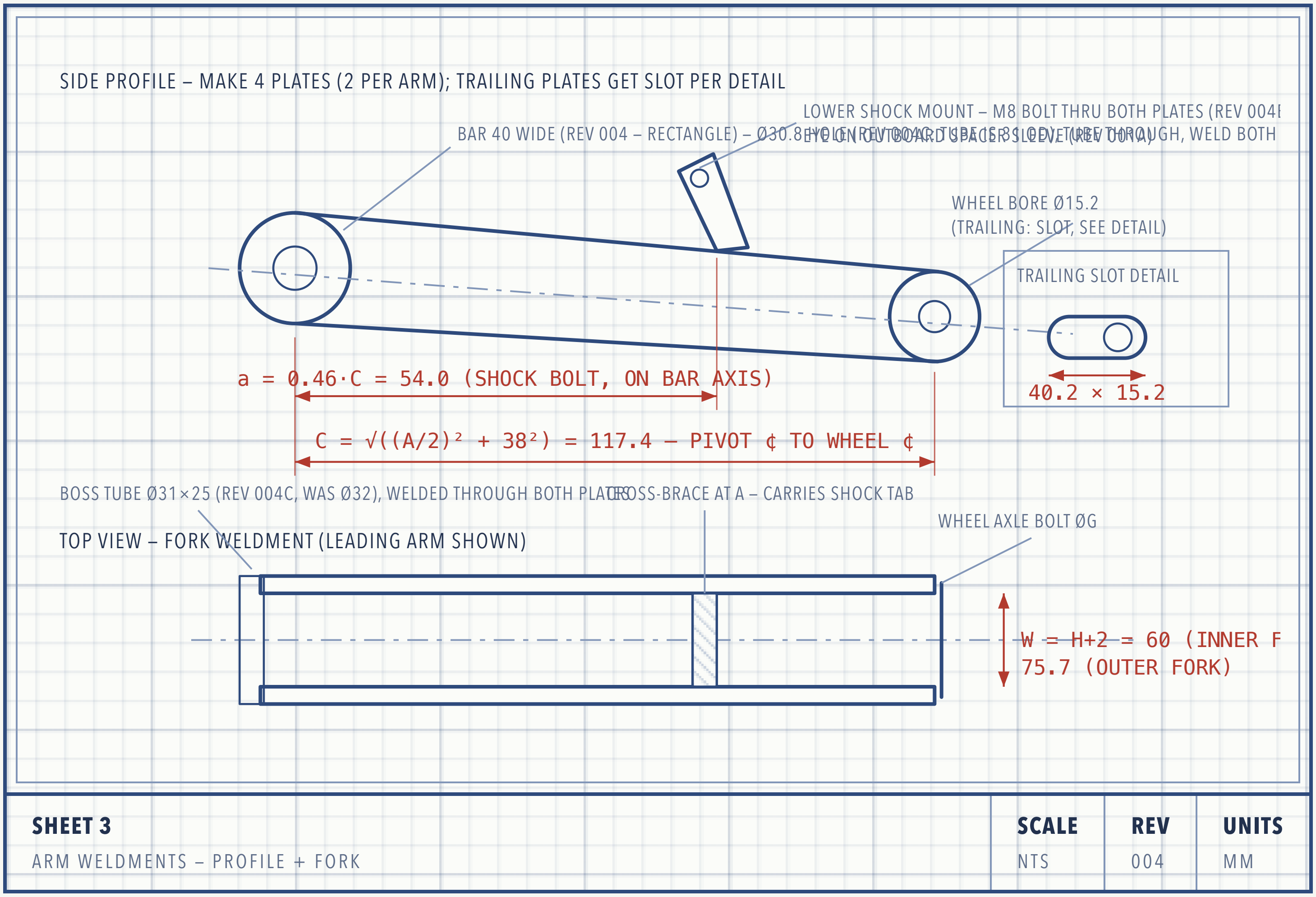
Both arms share one axle. To keep each idler wheel centred on the belt, each arm is a **fork** (two plates): the trailing arm is the narrow inner fork, the leading arm the wider outer fork, nested like scissors. Four bronze bushings — one pressed into each fork plate — carry the rotation. The axle itself is a partially-threaded M16 class-10.9 bolt clamping nothing: the nylock nut is set to leave running clearance, then the tube stack takes the clamp load. Rev 003a: each fork plate carries a 25-long boss tube — trailing bosses point *inboard*, leading bosses *outboard* (at the scissor interface the plates pass with only a 1.5 mm washer gap), so the bosses, centre spacer (≈14), two outboard sleeves (≈7) and six thrust washers form one continuous chain carrier-to-carrier.



- Bushing seats in the fork plates: bore Ø22 H7. Press bushings with a vise and a socket, never a hammer. Ream to Ø16 sliding fit after pressing if they close up.
- Grease: lithium EP2 at assembly; add a cross-drilled grease passage + M6 zerk in each fork boss if you want serviceability (recommended — this joint lives in mud).
- Three tube pieces set the lateral positions (Rev 003a): the centre spacer (≈14) between the trailing bushing flanges, and two outboard sleeves (≈7) between the leading flanges and the carriers. Cut all three at dry-stack so each idler wheel sits dead-centre in the belt (check against datum F before welding anything).

SHEET 3 ARTICULATING ARMS – LEADING & TRAILING

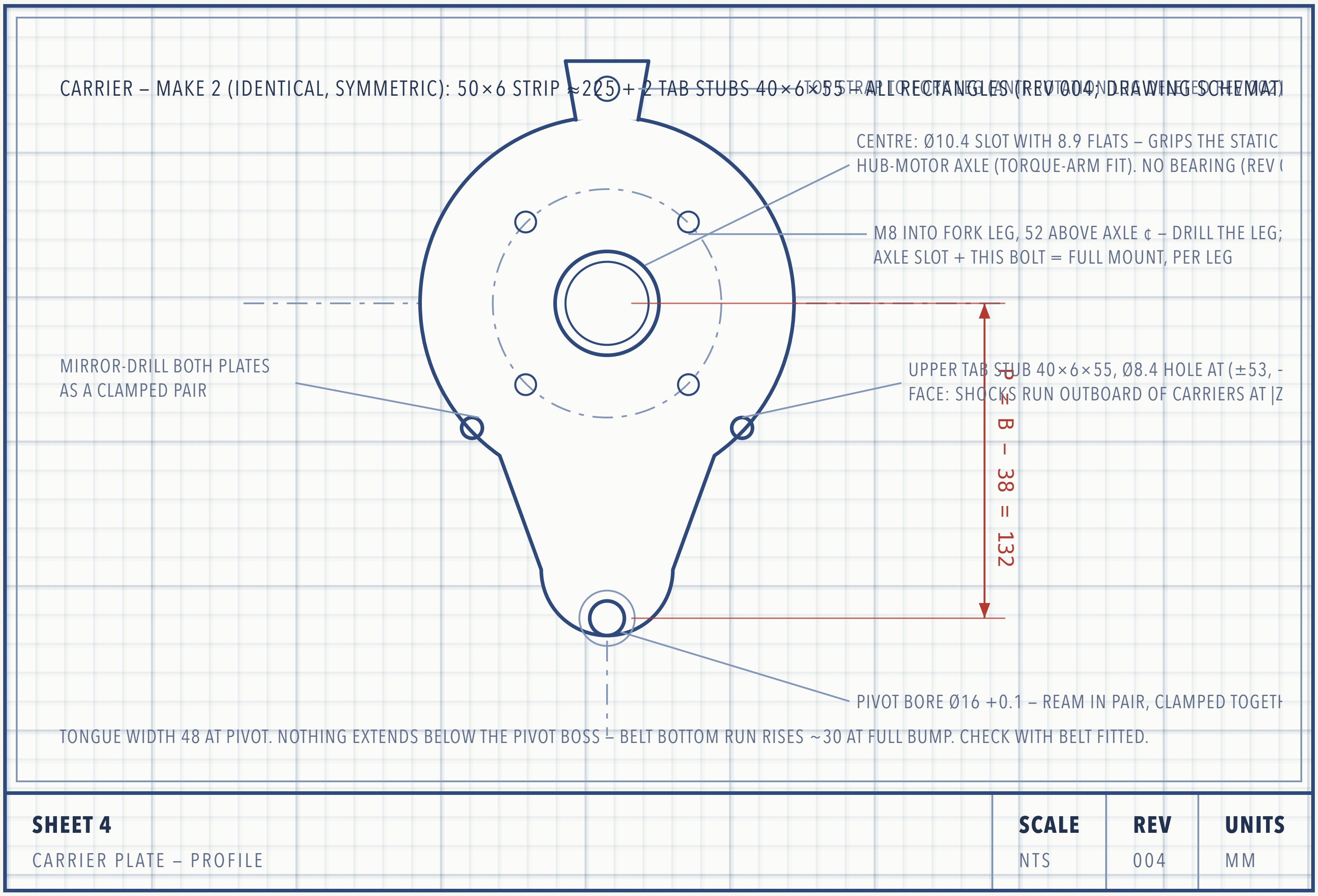
Each arm is a weldment of two plain **40 × 6.35 mm flat-bar plates** (Rev 004 — rectangles, straight cuts + drilled holes only: trailing bars ≈185, leading ≈166), a boss tube through each plate at the pivot end, and a cross-brace at 28–58 from the pivot. The two arms are identical *except*: the trailing arm's wheel bore is a **25 mm slot** for the belt tensioner (Sheet 6), and its fork is the narrower, inner one.



- **Arm length C is sacred.** At neutral (arms level) the wheel centres must land exactly where the stock frame held them, or the belt won't tension. Cut plates 1 mm long and file the wheel bores to final position with the belt fitted.
- Rev 004: each plate is a plain 40 mm-wide flat bar — plenty for a <150 kg vehicle corner. Don't drill lightening holes.
- Steel over aluminium: 6061-T6 loses ~40% strength in the weld HAZ and these arms are all welds. Use A36/S235 steel unless you're bolting, not welding.

SHEET 4 CARRIER PLATES – THE NEW BACKBONE

Rev 004: two **identical** flat-bar weldments — a **50 × 6 strip, cut ≈225**, plus two **40 × 6 × 55 tab stubs** lap-welded on the OUTER face at hub level (17 mm lap, clear of the axle slot) — hang from the scooter-fork legs, plates on the leg *outer* faces (|z| = 74 front AND rear — both gaps 140, legs 4 mm; Rev 003). Each has: a **Ø10.4 flatted axle slot** that grips the hub-motor axle like a torque arm, an **M8 hole 52 mm above** it that bolts into the drilled fork leg, the **Ø16 pivot bore 132 mm below** the axle, the **keel M8 at 104 mm below**, and shock-tab weld lobes reaching (±47, −5). The old centre bearing and anti-rotation lug are deleted — the fork does both jobs.



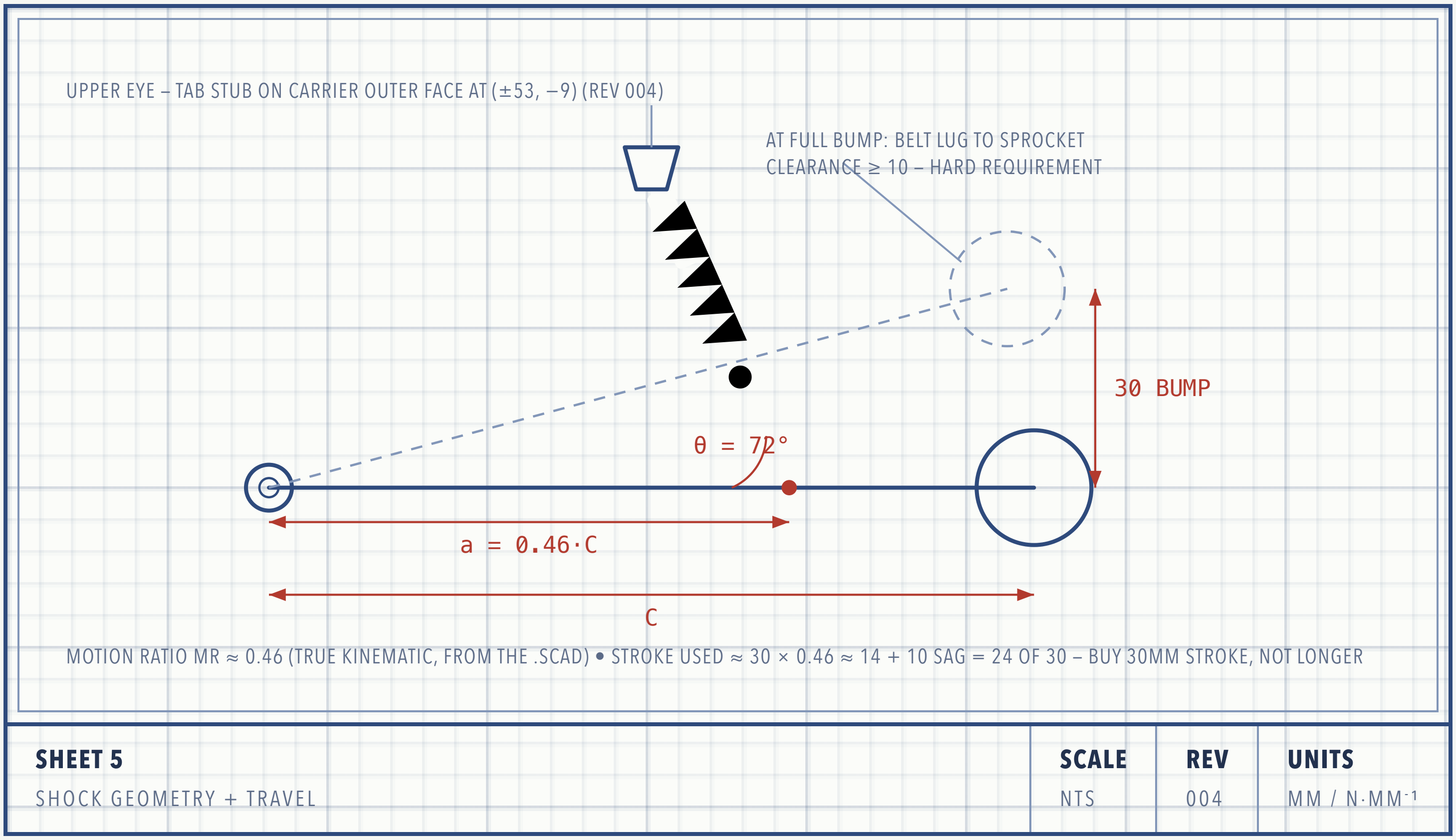
FIT-UP NOTE

The two carrier plates are joined into one rigid box by: the hub-motor axle (top), the pivot axle assembly (middle), and one M8 **keel standoff at 100 mm below the hub centre — between the sprocket's swept disc and the arm boss discs** (the 001a keel below the pivot would have hit the rising belt at full bump by ~40 mm; rev 003a slimmed the arm boss discs to Ø44 and the keel tube to Ø12×1.5 so this window truly exists: 4 mm to the sprocket, 4 mm to the boss discs, both articulation-invariant). Bolt both plates to the fork legs before drilling the keel holes so the box squares itself to the fork. The motor must spin freely with ≥3 mm side clearance each face.

The companion OpenSCAD model prints a **PASS/WARN belt-clearance line for every hanging part** (tongue, keel, pivot spacer, arm boss disc) at full bump, plus static gap checks for the keel window, the shock bolt and cross-brace vs the idler wheel, and the coil spring vs the carrier tongue — all must PASS with ≥5 mm before cutting steel, and the check must be re-run once the belt's guide-lug height is measured (datum F row).

SHEET 5 SHOCK GEOMETRY, TRAVEL & SPRING SIZING

The shock mounts at $a = 0.46 \cdot C$ along the arm — the lower eye sits **on the bar centreline** (rev 004, so a Ø10 hole fits the 40 mm flat bar) — inclined ~72° to it (near-vertical keeps the coil spring 6 mm off the carrier strip; rev 003a: the original 0.68·C through-bolt passed straight through the Ø108 idler wheel). Springs come out stiffer than the pre-003a numbers — recheck the worked method below before ordering.



SPRING RATE – WORKED METHOD

1. Weigh the corner: put the vehicle's wheel-position on a bathroom scale, loaded (rider + cargo). Call it M kg → force $F = 9.81 \cdot M$ N. That corner load splits across the pod's two arms roughly 50/50 → $F\_arm = F/2$.
2. Wheel rate target: static sag should be 25–30% of bump travel. Measured pod (C = 117.4) → bump travel 30 mm, sag ≈ 9 mm → $k\_wheel = F\_arm / 9$ N/mm.
3. Spring rate: $k\_spring = k\_wheel / MR^2$ with MR ≈ 0.46 → $k\_spring \approx 4.7 \times k\_wheel$.

Example — 160 kg loaded vehicle on 4 pods → M = 40 kg, F = 392 N, F_arm = 196 N → k_wheel ≈ 22 N/mm → **k_spring ≈ 104 N/mm ≈ 590 lb/in**. That's a heavy e-scooter / mini-moto spring.

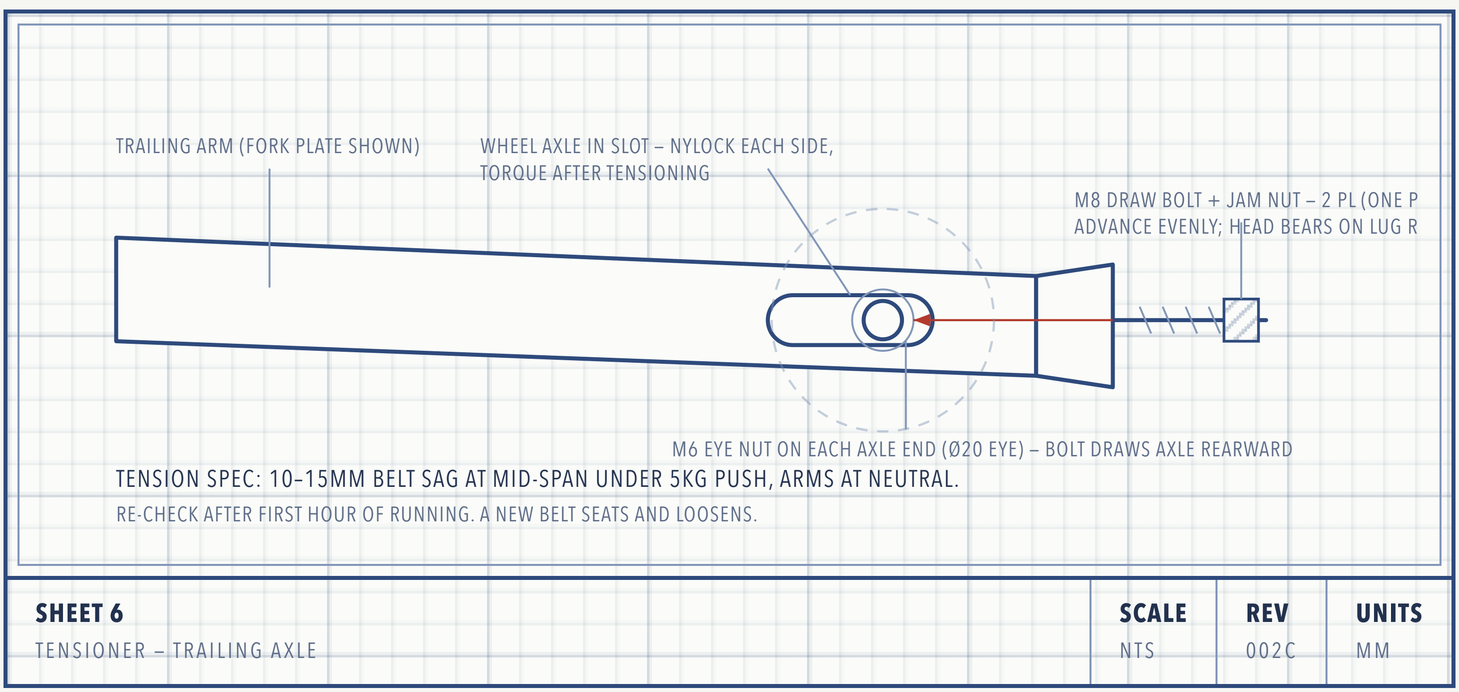
Buy adjustable-preload coil-overs, **150 mm eye-to-eye, 30 mm stroke (not longer — Sheet 5 stroke budget)**, and buy one spare spring one step stiffer and one softer. Weigh the real corner before ordering springs.

SHEET 6 BELT TENSIONER – TRAILING ARM

The articulating geometry cannot hold belt tension by luck. The mechanism is a **motorcycle chain adjuster** (Rev 002b — modeled explicitly in 3D): the trailing wheel axle rides in the Sheet-3 slot, and each protruding axle end carries an **M6 lifting eye nut** (DIN 582-style, Ø20 eye over the Ø15 axle — rev 004a: replaces the drilled-and-tapped block with a zero-fabrication catalog part). An **M6 draw bolt** passes through a clearance hole in a **lug welded to the fork-plate outer face** (front face 34 mm behind the nominal axle centre — behind the slot, forward of the plate tip) and threads into the block. The bolt head bears on the lug's rear face, so advancing it *draws the axle rearward* through the slot; belt tension, which pulls the axle forward, loads the bolt in tension, and a jam nut locks against the lug. Block-meets-lug is the natural stop at full ~25 mm take-up — enough to fit the belt slack and take up wear. See it in 3D: `render_mode="tensioner"` in the companion .scad (labeled close-up; `tension_pos` slides the axle 0–25).

WHY NOT A LUG AT THE PLATE TIP

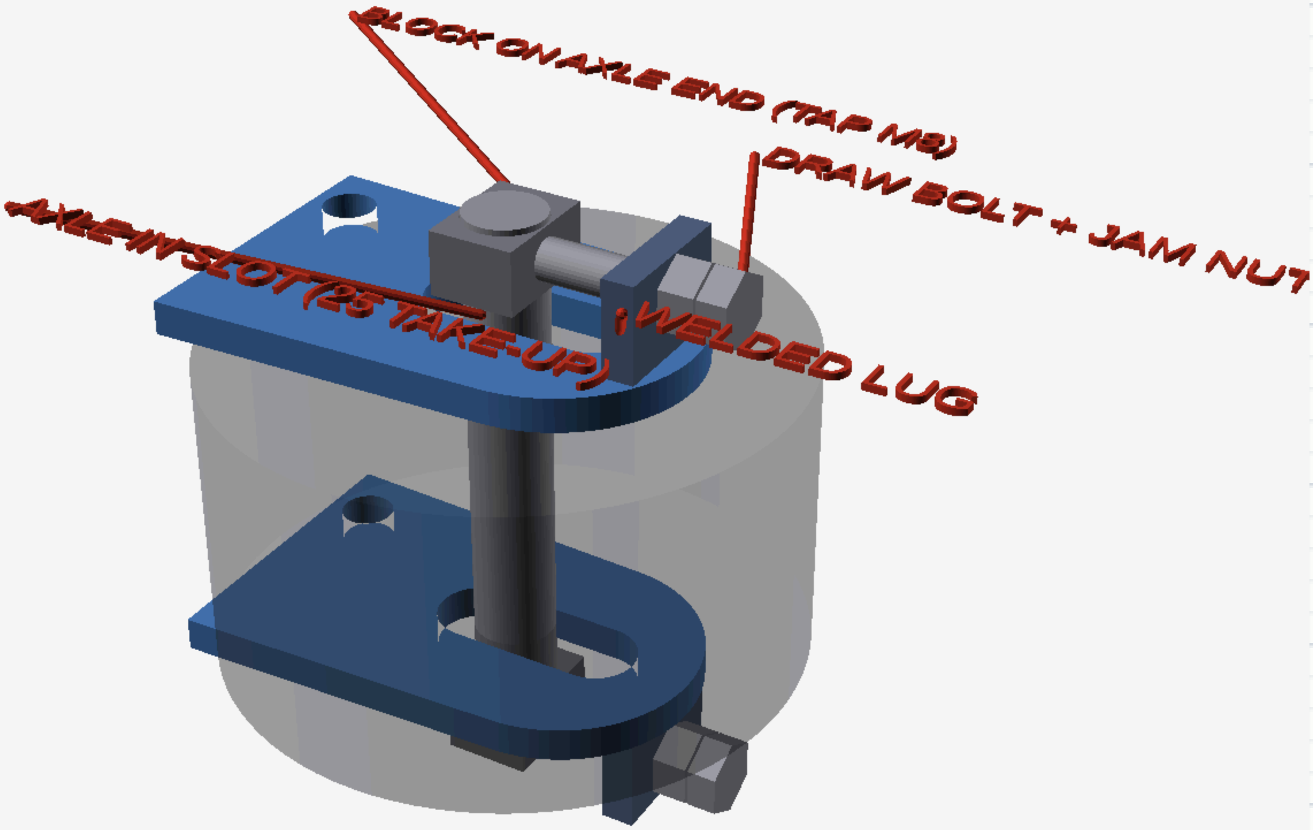
A lug at the very tip would put the bolt head 57–63 mm behind the axle at the slack position — **outside the belt's 54 mm wrap radius at the wheel, i.e. through the belt**. The 3D fit-check caught it; at the 34 mm lug station the head stays ≥2 mm inside the wrap at slack and ~10 mm at working tension. The .scad echoes a PASS/WARN line ("draw-bolt head to belt wrap") on every render.



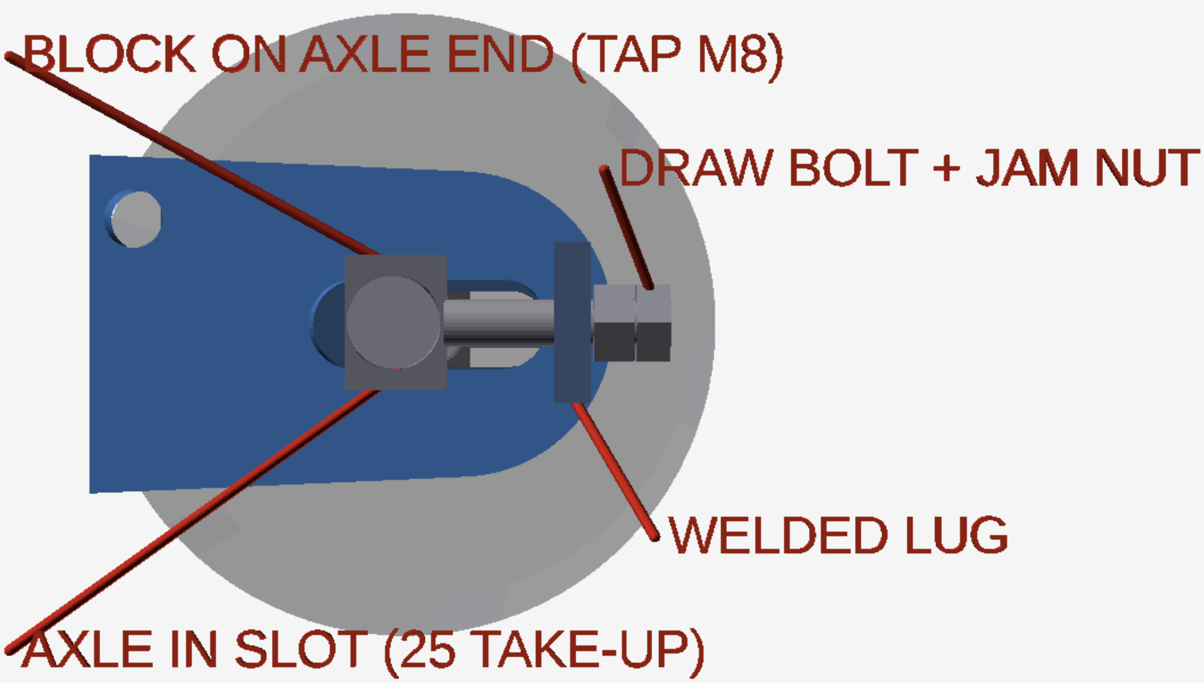
SHEET 6A – 3D DETAIL: HOW THE LUG, BOLT AND BLOCK CONNECT (RENDERS FROM THE .SCAD)

Step by step, this is how the belt gets tensioned — note the idler wheel itself never stretches; the *belt* is tensioned by dragging the wheel's axle rearward through the slot:

1. The trailing wheel's **Ø15 axle** sticks out past each fork plate and passes through the **40×15 slot**, so it can slide ~25 mm fore–aft.
2. An **M6 lifting eye nut** hangs on each protruding axle end (its Ø20 eye rides the Ø15 axle; the threaded boss faces the lug). Under load the eye pulls taut against the axle — the loose fit is harmless. Rev 004a: replaces the tapped block, zero fabrication.
3. The **welded lug** stands on the fork-plate outer face behind the slot (front face 34 mm behind the nominal axle centre), with a plain Ø8.5 clearance hole.
4. The **M6 draw bolt** passes through the lug's clearance hole and threads into the eye nut's boss (~8 mm engagement — fine for these tension-only loads). Its head bears on the lug's rear face — so turning the bolt clockwise *pulls the block, axle and wheel rearward*, taking the slack out of the belt. One bolt per side; advance both evenly, counting turns.
5. At 10–15 mm mid-span sag, torque the axle nylocks, then run the **jam nut** down against the lug so vibration can't back the bolt out.



BLOCK ON AXLE END (TAP M8)
DRAW BOLT + JAM NUT
AXLE IN SLOT (25 TAKE-UP)
WELDED LUG



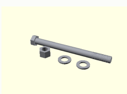
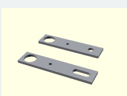
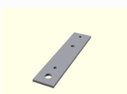
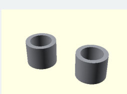
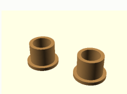
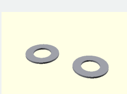
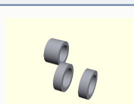
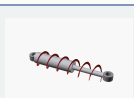
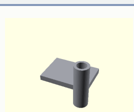
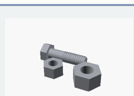
BLOCK ON AXLE END (TAP M8)
DRAW BOLT + JAM NUT
WELDED LUG
AXLE IN SLOT (25 TAKE-UP)

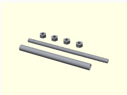
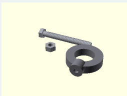
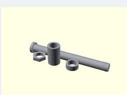
SHEET 6A	SCALE	REV	UNITS
TENSIONER – 3D DETAIL (RENDER_MODE="TENSIONER")	NTS	002C	MM

Renders from `apollo_track_pod.scad`, `render_mode="tensioner"` — the wheel is drawn translucent so the slot and axle stay visible; the far-side lug and bolt are visible at the bottom (one complete set per fork plate). Set `tension_pos` 0–25 in OpenSCAD to watch the axle travel through its take-up.

§7 BILL OF MATERIALS & SHOPPING GUIDE

The master table is the spec of record — every number on it comes from the Rev 004 sheets. Below it, shopping guides **§7.1–§7.9** turn each line into a purchase: exactly what to ask for, where to buy it, what an acceptable substitute is, roughly what it costs, and what to measure the day it arrives — *before* the return window closes. Order everything before cutting steel. Prices are 2026 US-dollar ballparks; verify at order time. The images are schematic 3D renders from the companion .scad (`render_mode="part"`) — they show the shape to look for, not a specific brand.

#	IMAGE	ITEM	QTY/POD	SPECIFICATION	NOTES / SOURCE	GUIDE
1		Pivot axle	1	`M16 × 195 cl.10.9, part-threaded, 45mm thread one end (both pods)`	Or Ø16 shaft + circlips; stack ≈ 182 both pods (gaps 140) — smooth shank 150 clears all bushings; only 1 nylock/bolt (has a head) — verify at dry-stack	§7.2
2		Arm bars	4	`40 × 6.35 (¼″) flat bar — trailing ≈185, leading ≈166`	Rev 004: straight cuts + drilled holes; trailing slot per Sheet 3	§7.1
3		Carrier strips + tab stubs	2+4	`50×6 strip ≈225 + 40×6 stubs ≈55`	Rev 004: all rectangles; ream pivot bores as a clamped pair	§7.1
4		Boss tubes	4	`Ø31 OD × Ø22 ID DOM tube, 25 long`	Bushing seats; one tube through each plate (Ø30.8 hole), welded both faces	§7.1
5		Bronze bushings	4	`SAE 841 flanged, Ø16 ID × Ø22 OD × 20`	Oil-embedded; press fit	§7.3
6		Thrust washers	6	`Ø16 × Ø30 × 1.5, hardened`	Between every rotating/static face pair	§7.3
7		Spacer + sleeves	1+2	`Ø22 OD × 3 wall — centre ≈14, outboard sleeves ≈7 ×2`	Close the pivot stack carrier-to-carrier — cut after trial stack-up	§7.1
8		Coil-over shocks	2	`150mm eye-to-eye, 30mm stroke, adj. preload`	Spring per Sheet 5 calc (~590 lb/in example). 30mm stroke max — longer lets the arm overshoot +15°	§7.4
9		Shock mounts	2+2	`2 tab stubs 40×6×55 w/ Ø8.4 hole (in BOM #3); 2 lower spacer sleeves Ø15 OD × ID 9 × ≈45`	Tab stubs lap-welded to carrier OUTER faces, eyes at (±53, −9) (rev 003a — shocks outboard at |z|=94); lower eyes on through-bolts + sleeves	§7.1/ §7.7
10		Fork mount hardware	2+2	`Hub-motor axle nuts M10 (stock) + M8×30 cl.10.9 + nylock per leg`	Replaces the 6205 carrier bearings — deleted rev 002 (fork-hung carriers; legs 4 mm)	§7.7

#	IMAGE	ITEM	QTY/POD	SPECIFICATION	NOTES / SOURCE	GUIDE
11		Keel standoff	1	Ø12×1.5 tube × 148 (both pods) + M8 rod cl.10.9 cut ≈ 172	At 100 below hub ¢, between sprocket swept disc and arm boss discs; boxes the carriers	§7.6
12		Tensioner draw bolts + blocks	2+2	M6×60 full-thread + M6 jam nuts; 2 M6 lifting eye nuts (DIN 582, eye Ø20 — ride the axle ends)	Sheet 6 (rev 004a) — eye nut replaces the tapped block; zero fabrication	§7.6
13	—	Anti-rotation link	—	DELETED rev 002	Carriers bolt straight to the fork legs — torque path is inherent	—
14		Hardware kit	—	Cl.10.9 (Gr8) bolts, nylock nuts, washers	Every fastener nylock or thread-locked; nothing plain-nut	§7.8
15		Grease fittings	2	M6 zerk + EP2 lithium grease	Optional but recommended, into fork bosses	§7.8
16		Idler axles	2	Ø15 — 15×100 MTB thru-axle (M15×1.5) or 15 mm shaft + collars	Idler bearings are 6302-2RS (15×42×13); ~90–120 axle length needed	§7.5
—		Re-used from stock pod	—	Hub-motor sprocket (Ø180, 10T), 2 idlers (Ø108×58), belt (1080×118), motor axle nuts	Idler bearings 6302-2RS — inspect; replace if rough	§7.9

§7.1 STEEL STOCK & CUT PLATES

BUY — BOM #2 · 3 · 4 · 7 · 9 · 11

WHAT TO BUY	**50 × 6 mm flat bar**, buy 0.5 m — the 2 carrier strips, cut ≈225 each (Sheet 4). **40 × 6.35 mm (1½″ × ¼″) flat bar**, buy 1.3 m — the 4 arm bars (trailing ≈185 ×2, leading ≈166 ×2; Sheet 3) and the 4 tab stubs (≈55 each; Sheet 4). **DOM tube Ø31 OD × Ø22 ID (4.5 wall)** — buy 120 mm, cut into four 25-long boss tubes (one per arm fork plate; kerf allowance). Rev 004c: owner sourced 31 OD (was 32) — still seamless/DOM only. **Tube Ø22 OD × 3 wall** — buy 400 mm; the pivot centre spacer (≈14) + 2 outboard sleeves (≈7) AND all the §7.5 idler-axle rings/sleeves (trailing 2×7 + ≈35, leading 2×8.8 + ≈50, per axle) are cut from it at dry-stack. **Tube Ø12 OD × 1.5 wall (ID 9)** — buy 250 mm; keel standoff cut to 148 (both pods — rev 003a: was Ø16; Ø12 is what fits the keel window). **Tube Ø15 OD × 3 wall (ID 9)** — buy 100 mm; 4 lower-shock spacer sleeves ≈45 long (rev 004b: ID 9 for the M8 bolts — shocks at |z|=94).
WHERE	Local steel supplier or any hardware store — flat bar is a stock shelf item (Rev 004: no cutting shop needed; every piece is a straight cut plus drilled holes). Online: OnlineMetals, Speedy Metals, Metals4U. The .scad `plates` mode is now a 1:1 drilling template — print it, glue it on, centre-punch.
SEARCH TERMS	`40x6 flat bar` / `1-1/2 x 1/4 steel flat bar` · `50x6 flat bar` / `2 x 1/4 flat bar` · `DOM tube 31mm OD 22mm ID` (≈ 1¼″ OD × 4.5 mm wall) — rev 004c, was 32mm OD
SUBSTITUTES	S235/S275 or plain hot-rolled mild steel for A36 — all fine. **Not aluminium** (Sheet 3 — welds lose ~40% strength in the HAZ). Boss tubes: seamless/DOM only, not thin-wall ERW. **Don't thin the carriers to 4 mm** even though the fork legs are 4: the flatted axle slot bears motor torque on a few mm² of edge and the pivot bore + shock-tab welds live in this plate. If 4 mm sheet is all you have, weld a 4 mm doubler ring around the axle slot and pivot bore (8 mm locally).
ROUGH COST	$15–35 of flat bar — no cutting-shop cost (Rev 004)
CHECK ON RECEIPT	Caliper the plate thicknesses (6.0 / 6.35); plates sit flat on a table with no rock; boss-tube ID reads *under* 22.0 — it must leave material to ream the bushing seats to Ø22 H7.

§7.2 PIVOT AXLE & M16 HARDWARE

BUY – BOM #1

WHAT TO BUY	1× **M16 hex bolt, part-threaded (owner sourced: 45 mm thread one end only)** per pod — length 195, same both pods (rev 004c; stack ≈182, smooth shank 150 clears all bushings with 4 mm margin). Plus 1× M16 nylock nut and 2× hardened Ø16 flat washers — **only one nut per bolt** (it has a head; do not buy 2 per pod).
WHERE	Fastener specialist (McMaster-Carr, Bel-Metric, Accu) or eBay/AliExpress.
SEARCH TERMS	`M16x195 partially threaded bolt` · `M16 nylock`
SUBSTITUTES	**M16 double-end stud × 190 + 2 nylocks** (owner plan): works like the part-threaded bolt with a nut each end — smooth middle section must be ≥150 so no thread reaches a bushing; prefer 8.8/B7 steel over 304 stainless (A2-70 is ~half the strength — still ~4× margin here, and use anti-seize on stainless threads). Or Ø16 ground shaft with circlip grooves + DIN 471 circlips — only if you can cut the grooves square (lathe).
ROUGH COST	$10–25
CHECK ON RECEIPT	**The smooth (unthreaded) shank must span all four bushings and the spacer** — bushings must never ride on threads. Confirm at the §9.3 dry-stack; if short, go longer and add washers under the head. Head must be stamped `10.9`.

§7.3 BRONZE BUSHINGS & THRUST WASHERS

BUY – BOM #5 · 6

WHAT TO BUY	4× **flanged bronze bushings, SAE 841 oil-embedded ("Oilite"), Ø16 ID × Ø22 OD × 20 long, flange ≈Ø28×3**. 6× **hardened thrust washers Ø16 × Ø30 × 1.5**.
WHERE	McMaster-Carr (flanged sleeve bearings), any bearing shop, AliExpress.
SEARCH TERMS	`flanged oilite bushing 16x22x20` · `thrust washer 16x30x1.5 hardened`
SUBSTITUTES	SAE 660 bearing bronze works (grease it more often). **No steel-on-steel and no plastic bushings** — this joint lives in mud under shock load.
ROUGH COST	\$3–7 per bushing · \$1–3 per washer
CHECK ON RECEIPT	Ø16 shaft/bolt slips through the bore and turns by hand (light drag is fine — seats are reamed after pressing); OD reads ≈22.05–22.10 for press-fit interference; flanges even, no cracks.

§7.4 COIL-OVER SHOCKS + SPRINGS

BUY – BOM #8

WHAT TO BUY	2× **coil-over shocks: 150 mm eye-to-eye, 30 mm stroke, Ø8 eye bores (owner-measured — drawing default was Ø10), adjustable spring preload**. Spring rate from the Sheet-5 worked method — weigh your real corner first; the worked example lands at ≈104 N/mm ≈ 590 lb/in (rev 004 — true kinematic MR ≈ 0.46 at the flat-bar shock station). Buy one spare spring one step softer and one stiffer.
WHERE	AliExpress / eBay — sold as mini-moto, pit-bike or e-scooter rear shocks.
SEARCH TERMS	`150mm rear shock absorber scooter` · `mini moto shock 150mm adjustable preload`
SUBSTITUTES	145–155 eye-to-eye is workable if you re-verify coil-bind and the +15° stop at fitting (§9.5). A longer *stroke* is **not** a substitute.
ROUGH COST	$15–45 each
CHECK ON RECEIPT	Eye-to-eye 150±2 (hole centre to hole centre); compress in a vise — full stroke ≤30; eye bore **confirmed Ø8** (owner measured, rev 004b): all shock pins/bolts are M8 and the sleeves Ø15 OD × ID 9, geometry unchanged; the preload collar actually turns. Note: spring-only (undamped) units are an accepted owner deviation from the oil-damped spec — see the §10 test protocol emphasis on belt-walk under repeated bumps.

DO NOT BUY

Longer-stroke shocks. 30 mm stroke is a hard ceiling (Sheet 5 stroke budget: ≈14 used + 10 sag = 24 of 30). More stroke lets the arm overshoot +15° — the belt walks off and the lugs hit the sprocket.

§7.5 IDLER AXLES

BUY – BOM #16

WHAT TO BUY

2× **Ø15 axles, 90–120 long**. Cleanest off-the-shelf part: a **15×100 MTB thru-axle (M15×1.5)**. Otherwise Ø15 ground shaft + shaft collars and nylocks. *Owner note (rev 004a): **MEROCA 15×148 alloy thru-axles ordered** — Ø15 shank, 148 overall, only a **9 mm M15×1.5 thread at the tip**, so the spacer stack must place the nut exactly on that thread. Add 4× **M15×1.5 nuts** (fine pitch — order online) + ~8 Ø15 washers; all sleeves below cut from the §7.1 Ø22×3 tube (ID 16 slides over the shank), cut-to-fit at dry assembly (start long, shorten until the nut seats with 1–2 threads proud).*

***TRAILING axle stack** (head → tip): head → eye nut (8) → standoff ring ≈7 (sets the eye at the lug-hole height) → plate → washer Ø15×1 → wheel (bearing · inner tube · bearing) → washer → plate → standoff ≈7 → eye nut → **filler sleeve ≈35** → washer → M15×1.5 nut.*

***LEADING axle stack**: head → plate → **centering sleeve ≈8.8** (the leading fork is 75.7 inner vs the 58 wheel — sleeves, not washers) → wheel → sleeve ≈8.8 → plate → **filler sleeve ≈50** → washer → M15×1.5 nut.*

Alloy is acceptable — verify the shank is a true Ø15.0 with no slop in a 6302 on arrival. Plus, per axle: **2 spacer washers Ø15 bore × ~1–2 thick** (fork plate → bearing INNER race) and an **inner spacer tube between the two 6302 inner races** inside the hub — the stock wheel very likely already has these; reuse the donor pod's axle spacers if present. The clamp path must be plate → washer → inner race → tube → inner race → washer → plate, so the axle can be torqued without the plates pinching the spinning hub (which runs 1 mm clear of each plate face).

WHERE

Bike shop, Amazon, AliExpress.

SEARCH TERMS

`thru axle 15x100` · `15mm ground shaft`

SUBSTITUTES

Any Ø15 high-tensile shaft that fills the 6302-2RS bore; don't shim a Ø12 bolt up to 15, and **not a threaded stud/rod** — M14 shank (Ø14) wobbles 1 mm in the bearing, M16 (Ø16) won't enter it; bearings need a true Ø15 seat.

ROUGH COST

$8–20 each

CHECK ON RECEIPT

Slides through a stock idler bearing (6302-2RS, Ø15 bore) without force and without slop; long enough to cross the arm fork (inner width 60 /

75.7) plus both plates and the nut. With the wheel + spacers stacked on it, torquing the axle must NOT drag the wheel — if it does, the inner spacer tube is missing or short.

§7.6 TENSIONER & KEEL HARDWARE

BUY – BOM #11 · 12

WHAT TO BUY
2× **M6×60 full-thread draw bolts** (cl.8.8 or better) + 2 M6 jam nuts — the Sheet-6 tensioner (rev 002b mechanism, rev 004a hardware). 2× **M6 lifting eye nuts, DIN 582-style, eye inner Ø20** (owner has these) — the eye rides the protruding Ø15 axle end; no drilling, no tapping. 1× **M8 threaded rod, class 10.9** (sold as 1 m sticks — cut to ≈172, both pods) + 4 nuts + washers for the keel standoff. The Ø12×1.5×148 keel tube itself is in the §7.1 steel order.

WHERE
Any fastener supplier or hardware store with a graded-fastener aisle; shaft collars from a bearing shop or Amazon.

SEARCH TERMS
`M8 threaded rod class 10.9` · `M6x60 full thread 8.8` · `M6 eye nut DIN 582` · `15mm shaft collar`

SUBSTITUTES
Grade B7 or Grade 8 rod for the keel. Not unmarked hardware-store rod — the keel is one of the three members boxing the carriers.

ROUGH COST
Under $15

CHECK ON RECEIPT
Rod is marked/spec'd high-tensile; draw bolts run freely through their jam nuts; blocks/collars slide on a Ø15 axle without slop.

§7.7 FORK-MOUNT & SHOCK HARDWARE

BUY – BOM #9 · 10

WHAT TO BUY	2× **M8×30 cl.10.9 bolts + nylocks** — one per fork leg (carrier 6 + leg 4 + nut; rev 002c). 2× **M8 part-threaded bolts + nylocks** — lower shock through-bolts (rev 004b: eyes measured Ø8); length to suit the arm fork + sleeve, measured at §9.5 fitting (≈135). The stock hub-motor **M10 axle nuts are reused**. The Ø15 OD × ID 9 × 45 spacer sleeves are in the §7.1 steel order; the 2 upper shock tabs are cut from the §7.1 6 mm plate.
WHERE	Fastener supplier.
SEARCH TERMS	`M8x30 10.9 bolt` · `M8 partially threaded bolt 10.9 140`
SUBSTITUTES	Grade 8 (US) equals class 10.9 here. Nothing below 8.8 anywhere on the pod.
ROUGH COST	A few dollars
CHECK ON RECEIPT	The M8 shock bolts have smooth shank where the eye and sleeve ride; all heads carry a grade stamp.

§7.8 CONSUMABLES & SHOP SUPPLIES

BUY – BOM #14 · 15 + §8

WHAT TO BUY	Nylock nut + hardened washer assortment, M8/M10, class 10 / Gr8. 2× **M6 grease zerks** + EP2 lithium grease (+ grease gun). MIG wire **ER70S-6** 0.8 mm (+ shielding gas per your welder). **Zinc-rich primer** + topcoat, degreaser, masking tape. Medium (blue) threadlocker · **paint pen** for torque witness marks. **Ø22 H7 reamer** (bushing seats) and a Ø16 hand/adjustable reamer (bushings after pressing) — borrow or rent if you don't own them; step drills for the wheel/shock bores, and a Ø31 step bit or bi-metal hole saw (rev 004c, was 32 — Ø31 saws are less common; Ø30 + hand-ream/file 0.8 mm oversize also works) for the four Ø30.8 pivot holes (rev 004c, was 31.8; hand-drilled on a drill press — the boss tube passes through and the both-face weld fills the fit).
WHERE	Welding supply + hardware store; reamers from a tool supplier (or any local machine shop can ream the seats for beer money).
ROUGH COST	$40–80 if starting from nothing (reamers extra)
CHECK ON RECEIPT	Zerk thread is M6; primer can actually says zinc-rich (weld-through or cold-galvanising), not plain grey primer.

§7.9 REUSED PARTS & RECEIVING INSPECTION

REUSE – NO PURCHASE

From the donor pod, reuse: the **hub-motor sprocket** (Ø180, 10T), both **idler wheels** (Ø108×58 on 6302-2RS bearings), the **belt** (1080×118, 18 links × 60 mm pitch) and the **M10 motor-axle nuts**. Spin every 6302-2RS by finger: any grit, roughness or wobble → replace (search `6302-2RS bearing`, $2–4 each — buying 4 spares up front is cheap insurance).

Same-day inspection table — run it the day the parts arrive, while returns are still easy:

PART	MEASURE ON ARRIVAL	MUST READ
Plates (§7.1)	Thickness, flatness	`6.0 / 6.35 · flat, no rock`
Boss tube (§7.1)	Inside diameter	`< 22.0 (ream to Ø22 H7)`
Bushings (§7.3)	OD · ID	`≈22.05–22.10 · Ø16 slips, turns by hand`
Thrust washers (§7.3)	Bore · thickness	`Ø16 · 1.5`
Pivot bolt (§7.2)	Smooth-shank length · head stamp	`Shank spans all 4 bushings · "10.9"`
Shocks (§7.4)	Eye-to-eye · stroke · eye bore	`150±2 · ≤30 · Ø8 (confirmed — M8 bolts)`
Idler axles (§7.5)	Diameter · length	`Ø15 slip in 6302-2RS · 90–120`
Keel rod (§7.6)	Grade marking	`10.9 / B7 / Gr8`
Idler bearings (reused)	Finger-spin	`Smooth, no grit — else replace 6302-2RS`

§ 8 FABRICATION SPECIFICATIONS

CUTTING	Rev 004: every plate is a flat-bar rectangle — chop saw or angle grinder for the straight cuts, drill press for every bore. Finish cut edges to bright metal; no hand drilling of bearing/bushing seats.
CRITICAL BORES	Bushing seats Ø22 H7, reamed in the welded boss tubes (the plate itself gets a drill-press Ø30.8 hole — rev 004c, was 31.8 — step bit or hole saw — that the Ø31 tube passes through). Pivot bores in carrier plates reamed as a clamped pair so the axle is square. Wheel bores ØG +0.2.
WELDING	MIG, ER70S-6, 5 mm fillets. Boss tubes welded both faces. Shock tabs welded both sides — a single-side tab weld is the classic failure. No preheat needed on A36 at these thicknesses. Weld with bushings *out* (heat kills them); press bushings after paint.
SQUARENESS	Fork plates must be parallel within 0.5 mm over their length, or the idler runs angled and throws the belt. Tack, measure, weld in a staggered sequence, re-measure.
FINISH	Degrease, zinc-rich primer, topcoat. Mask every bore and bearing seat.
TORQUE	M8 cl.10.9 — 30 N·m · M10 — 60 N·m · M12 — 105 N·m · M16 pivot nylock — snug to zero end-float then back ⅛ turn (running fit, not clamped). Mark every fastener with paint-pen witness lines.

§9 BUILD INSTRUCTIONS & ASSEMBLY

§9.0 is the master sequence — the order of operations, unchanged from Rev 002. Chapters **§9.1–§9.8** then expand every component into a full build page: the tools you need, numbered steps, the checks that must pass before you move on, and the classic mistakes. Nothing in the chapters overrides a sheet dimension — when in doubt, the drawing wins.

§9.0 MASTER SEQUENCE

1. **STRIP THE DONOR POD.** Remove belt, idlers, sprocket. Photograph and measure the internal frame and anti-rotation feature (datum J) before discarding it.
2. **PRESS BUSHINGS.** Four flanged bronze bushings into the fork-plate bosses (vise + socket). Verify the Ø16 axle turns by hand with no rock. → §9.3
3. **DRY-STACK THE PIVOT.** Thread onto the axle: carrier L → outboard sleeve → washer → leading fork → washer → trailing fork → washer → centre spacer → washer → trailing fork → washer → leading fork → washer → outboard sleeve → carrier R (Sheet 2 order, rev 003a). Cut the centre spacer + both sleeves so each idler centreline lands mid-belt (datum F) and the stack is snug carrier-to-carrier. No welding yet. → §9.3
4. **HANG AND BOX THE CARRIERS.** Slot both plates onto the hub-motor axle, bolt the M8 into each drilled fork leg, then fit the keel standoff (drill its holes with the plates bolted up so the box squares to the fork). Spin the motor — ≥3 mm clearance each face, zero rub, and check the belt's 11 mm/side gap to the fork legs. → §9.4
5. **LOCATE SHOCK MOUNTS.** With arms level and shocks held in place at ~30% compression, mark positions: upper tab stubs on the carrier *outer* faces with eyes at (±53, −9), lower eyes on Ø10 through-bolts with spacer sleeves at a = 54.0 on the bar axis, θ = 72° (shocks run outboard of the carriers at |z| = 94 — rev 004). Tack the tabs, articulate ±15° by hand checking nothing binds, then final-weld with bushings removed. → §9.5
6. **PAINT, THEN FINAL-ASSEMBLE** the pivot stack with grease, set the M16 nylock for free running, fit shocks. → §9.8
7. **FIT IDLERS** to arm forks — leading in the round bore, trailing in the slot, jack bolts backed off fully. → §9.6
8. **WRAP THE BELT.** Sprocket first, engage lugs, lever over the idlers with the trailing axle at its most-forward slot position (rubber levers only — no screwdrivers into the carcass). → §9.7

9. **TENSION.** Advance both jack bolts evenly to 10–15 mm mid-span sag, square the axle in the slot (equal bolt counts), torque the axle nylocks. → §9.6
10. **FINAL TORQUE.** Motor axle nuts + M8 fork bolts + pivot nylock witness-marked, then run the §10 test protocol before carrying a rider. → §9.8

Cutting the bars (§9.1, §9.2) happens in parallel with steps 1–3 — all straight cuts from stock flat bar (Rev 004), no waiting on a cutting shop.

§9.1 CARRIER PLATES – CUT, DRILL, FIT

Two identical 6 mm plates (Sheet 4) — the new backbone. Everything else hangs off these, so the holes must match between the two plates exactly.

Tools: drill press (no hand drilling of bores), step drills, Ø16 reamer, round file, clamps, calipers, square, angle grinder if hand-cutting.

1. **CUT THE CARRIER STRIPS.** Two 50 × 6 flat-bar strips at ≈225, ends square. Print the .scad `plates` layout 1:1 as a drilling template (or mark from the Sheet-4 dimensions), glue it on, centre-punch every hole.
2. **CLAMP THE TWO PLATES TOGETHER** perfectly aligned and keep them clamped for every hole. Sheet 4: *mirror-drill both plates as a clamped pair* — this is what makes the pivot axle sit square.
3. **DRILL THE PIVOT BORE.** Step-drill to Ø15.5, then ream to **Ø16 +0.1** with the pair still clamped. Never hand-drill this bore.
4. **FINISH THE AXLE SLOT.** The centre slot is **Ø10.4 with 8.9 flats** — file until it slides snugly over the flatted hub-motor axle with *no rock*. This slot is the torque arm; a sloppy fit hammers itself loose.
5. **DRILL THE M8 FORK-LEG HOLE** 52 mm above the axle centre. Cut the four 40 × 6 tab stubs (≈55) and drill each one's Ø8.4 eye hole (M8 pin — eyes measured Ø8, rev 004b) so it lands at (±53, −9) from the hub centre once the stub is lap-welded on the OUTER face at hub level (17 mm lap onto the strip, clear of the axle slot, ABOVE the coil-spring corridor).
6. **(THE TAB STUBS ABOVE ARE THE UPPER SHOCK MOUNTS** — Ø8.4 holes, lap-welded to the carrier OUTER faces at §9.5; don't weld them yet.)
7. **DEBURR EVERYTHING**, break edges 0.5 mm, both sides of every bore.

CHECK BEFORE MOVING ON

Stacked together, the two plates match edge-to-edge and every hole lines up with a pin through it. The Ø16 axle passes through both pivot bores and turns without binding. The axle slot grips the motor-axle flats snugly. The keel holes are *not* drilled yet — they're drilled on the scooter at §9.4.

COMMON MISTAKES

Drilling the plates separately (pivot bores misalign → axle binds → arms stick). Hand-drilling the pivot bore oversize. Filing the axle slot loose "so it goes on easier" — the torque-arm fit must be snug. Drilling the keel holes on the bench instead of bolted to the fork.

§9.2 ARM WELDMENTS – LEADING & TRAILING

Each arm is a fork of two 6.35 mm plates + boss tube + cross-brace + shock tab (Sheet 3). The two arms are identical except the trailing arm has the 40.2×15.2 tensioner slot instead of a round wheel bore, and its fork is the narrower inner one (60 vs 75.7).

Tools: MIG welder + ER70S-6, angle grinder, drill press, Ø22 H7 reamer, flat welding table, machinist square, spacer blocks, calipers, clamps.

1. **CUT THE FOUR ARM BARS** from 40 × ¼″ flat bar per Sheet 3 (trailing ≈185, leading ≈166) — **cut 1 mm long at the wheel end**. Arm length C = 117.4 is sacred: the extra millimetre gets filed to final position with the belt fitted, never guessed.
2. **MACHINE THE WHEEL ENDS.** Leading-arm plates: round bore Ø15.2. Trailing-arm plates: slot 40.2 × 15.2 per the Sheet 3 detail. Drill-press + file; keep the slot sides parallel.
3. **PREPARE THE BOSS.** Drill the Ø30.8 pivot hole (rev 004c, was 31.8; step bit or hole saw, drill press only); slide the Ø31 boss tube through (rev 004c, was 32 — light interference/tap fit), end flush with the plate face per Sheet 2 — trailing plates flush at the OUTER face, leading plates at the INNER face — and weld it on *both* faces (§8). The tube's Ø22 bore is the bushing seat — ream to **Ø22 H7** after welding (welding distorts it slightly; ream after, not before).
4. **JIG THE FORK.** Stand the two plates of one arm on the flat table at the fork inner width — **60 (trailing) / 75.7 (leading)** — using spacer blocks, bosses aligned on a Ø16 bar through both bushless seats.
5. **FIT THE CROSS-BRACE AT 28–58 FROM THE PIVOT CENTRE** (rev 003a: inboard of the wheel — the old a±25 station passed through the Ø108 idler), and confirm the Ø10 shock-bolt holes sit at a = 54.0 on the bar centreline. Tack only.
6. **MEASURE, THEN WELD STAGGERED.** Fork plates must be parallel within **0.5 mm** over their length (§8 — an angled idler throws the belt). Tack → measure → short staggered welds alternating sides → re-measure. 5 mm fillets.
7. **WELD THE SHOCK TAB ON BOTH SIDES.** §8: a single-side tab weld is the classic failure.

CHECK BEFORE MOVING ON

Pivot-to-wheel centre reads C = 117.4 (+1 mm filing allowance) on all four plates. Fork plates parallel ≤0.5 mm. A Ø15 bar through the wheel bore/slot sits square to a Ø16 bar through the boss. Bushings are *not* pressed yet — §8: weld with bushings out, press after paint.

COMMON MISTAKES

Welding with the bronze bushings installed (heat kills them). One long continuous weld per side — warps the fork out of parallel. Single-side shock-tab welds. Cutting the plates to exact length and then discovering the belt won't tension — cut long, file to fit. Drilling lightening holes: don't, on the first pod.

§9.3 PIVOT STACK – BUSHINGS, DRY-STACK, SPACER

Both arms swing on one M16 axle (Sheet 2). The stack is built *dry* first to cut the spacer and verify the axle; final greased assembly happens after paint (§9.8).

Tools: bench vise, large sockets (press tooling), calipers, tube cutter or saw, Ø16 reamer, EP2 grease for final assembly.

1. **PRESS THE BUSHINGS** — one flanged bronze bushing into each of the four fork bosses. Vise + a socket that bears on the bushing's outer edge, flange toward its thrust washer — **trailing-fork flanges face inboard (toward the centre spacer), leading-fork flanges face outboard (toward the sleeves)**. Never a hammer.
2. **VERIFY THE FIT.** The Ø16 axle must turn by hand in every bushing with no rock. If a bushing closed up in pressing, ream it to a Ø16 sliding fit.
3. **DRY-STACK IN SHEET 2 ORDER (REV 003A):** carrier L → **outboard sleeve** → washer → leading fork (boss pointing out) → washer → trailing fork (boss pointing in) → washer → **centre spacer** → washer → trailing fork → washer → leading fork → washer → **outboard sleeve** → carrier R. Thrust washer between every rotating/static face pair — 6 per pod.
4. **CUT THE SPACER SET.** Three Ø22 × 3-wall pieces (centre ≈14, outboard sleeves ≈7 ×2) set where both forks sit sideways AND close the stack to the carriers. Measure against datum F so **each idler wheel lands dead-centre in the belt**, then cut square and deburr. This is why the tube ships long.
5. **CHECK THE AXLE LENGTH.** ≈215 (front pod) / 235 (rear): full nut engagement, smooth shank spanning all four bushings — threads never inside a bushing.
6. **SET THE NUT — RUNNING FIT.** At final assembly (§9.8, greased): M16 nylock snug to zero end-float, then **back off ⅛ turn**. The axle clamps nothing — the spacer takes the load; the forks must swing free.

> CHECK BEFORE MOVING ON
>
> Both arms swing through their full arc by hand — smooth, no binding, no lateral rock. Idler centrelines sit mid-belt per datum F. Drill and tap the M6 zerk into each fork boss now if you're fitting grease passages (recommended — this joint lives in mud).

COMMON MISTAKES

Hammering bushings in (cracks the flange, ovals the bore). Cutting the spacer set to the drawing's ≈14/≈7 without dry-stacking — it's a fit dimension, not a spec. Torquing the M16 like a normal bolt and crushing the forks: it's a running fit, snug minus ⅛ turn. Forgetting a thrust washer — every steel face that rotates against another needs one.

§9.4 HANGING THE CARRIERS & FITTING THE KEEL

The two carriers plus the hub axle, pivot assembly and keel standoff form one rigid box. The box must square itself to the scooter fork — which is why the keel holes are drilled *in place*, last.

Tools: drill + Ø8.5 bit, transfer punch, wrenches, feeler gauges or 3 mm shim.

1. **SLOT THE PLATES ONTO THE MOTOR AXLE.** Both carriers slide over the flatted Ø10 hub-motor axle, sitting on the fork-leg *outer* faces (|z| = 74, both pods — gaps 140, legs 4 mm). Stock M10 axle nuts go back on, finger-tight for now.
2. **DRILL THE FORK LEGS.** Using each plate's M8 hole (52 above the axle centre) as the template, mark, remove plate, drill the leg Ø8.5, refit, and bolt **M8×30 + nylock** through plate and leg.
3. **DRILL THE KEEL HOLES IN PLACE.** With both plates bolted up tight, transfer-drill the keel holes at **100 below the hub centre** through both plates. Drilled bolted-up, the box squares to the fork automatically.
4. **FIT THE KEEL STANDOFF:** Ø12×1.5 tube ×148 (both pods) over the M8 rod (cut ≈172), washers and nuts each end, snugged so the tube is in compression between the plates. It must sit in the window *between the sprocket's swept disc and the Ø44 arm boss discs* — 4 mm to the sprocket, 4 mm to the boss discs (rev 003a).
5. **SPIN THE MOTOR BY HAND.** Free rotation, **≥3 mm side clearance each face**, zero rub anywhere.
6. **BOTH PODS:** confirm the 118 belt runs centred in the 140 fork gap — **11 mm per side** (rev 003; the old 1 mm/side front-fit watch-item is retired).

CHECK BEFORE MOVING ON

Box is rigid — grabbing one carrier and pushing does not lozenge the pair. Motor spins free with ≥3 mm/face. Pivot axle through both carrier bores turns freely (bores still aligned after bolt-up). Keel clears the sprocket's swept circle and the arm boss discs by ~4 mm each, at every arm angle.

COMMON MISTAKES

Drilling the keel holes on the bench — the box then fights the fork and the pivot binds. Putting the keel below the pivot (the deleted 001a position — it collides with the belt at full bump). Over-torquing the stock axle nuts before the M8 legs bolts are in, twisting the plates.

§9.5 SHOCK MOUNTS & INSTALLATION

Shock geometry (Sheet 5): lower eye on the arm at a = 0.46·C = 54.0, on the bar centreline (rev 004; the pre-003a 0.68·C bolt sat inside the idler wheel), upper eye on a tab stub on the carrier *outer* face at (±53, −9), shock inclined θ = 72° — shocks run *outboard* of the carriers at |z| = 94 (rev 003: with 140 gaps the carriers sit at |z| = 74, 15 mm outside the belt edge — still too little for an inboard shock, so the rev-002 inboard placement stays retired). Position is everything here; weld last.

Tools: MIG welder, clamps, angle gauge or protractor, marker, an assistant (or zip ties) to hold shocks at length.

1. **ASSEMBLE THE PIVOT TEMPORARILY** (dry stack from §9.3) with both arms on, on the hung carriers from §9.4. Set both arms level — neutral.
2. **HOLD EACH SHOCK AT ~30% COMPRESSION** just outboard of the carrier: lower eye at the shock-bolt station 54.0, body leaning at θ = 72° to the arm, upper eye landing at (±53, −9) from the hub centre.
3. **MARK AND TACK THE UPPER TABS** on the carrier *outer* faces at that spot. Tack only — no full welds yet.
4. **FIT THE LOWER THROUGH-BOLTS:** M8 part-threaded bolt (rev 004b — eyes measured Ø8) through both arm plates, shock eye riding on the Ø15 OD × ID 9 × ≈45 spacer sleeve reaching out past the carrier plane, nylock. Measure the bolt length you actually need here (§7.7, ≈135).
5. **HAND-CYCLE THE SUSPENSION ±15°** — full droop to full bump, several times. Nothing may bind; the shock body must clear the carrier edge and the sprocket at every angle; no coil-bind at full bump (stroke used ≈27 of 30, Sheet 5).
6. **FINAL-WELD THE TABS** — both sides of every tab (§8), with the bronze bushings *removed* from the arms and the shocks off. Let cool, refit, re-cycle ±15° to confirm nothing moved.

CHECK BEFORE MOVING ON

At full bump: belt-lug-to-sprocket clearance ≥10 mm (Sheet 5 hard requirement) and the shock is not coil-bound. At full droop the shock still holds the belt path taut (spring preload — §0). Both sides measure the same eye-to-eye at neutral.

COMMON MISTAKES

Final-welding the tabs before hand-cycling — if the geometry is off you've welded a bind into the carrier. Welding with bushings or shocks in place. Mounting the upper tabs on the carrier *inner* face — that was Rev 002 with thick legs; with the measured 4 mm legs there is no inboard room (Rev 003 runs shocks outboard, |z| = 94). Fitting longer-stroke shocks "for more travel" — the arm overshoots +15° and throws the belt.

§9.6 BELT TENSIONER – IDLERS, DRAW BOLTS, TENSIONING

The trailing wheel axle rides in the arm's 40.2×15.2 slot and is **drawn rearward by two M6 draw bolts** reacting against lugs welded to the fork-plate outer faces (Sheet 6, rev 002b — a motorcycle chain adjuster: bolt through the lug, threaded into a tapped block on the axle end, head + jam nut on the lug's rear face). ~25 mm of range to fit the belt and take up wear. See it in 3D: `render_mode="tensioner"` in the .scad.

Tools: MIG welder (lugs), drill + M8 tap (blocks), wrenches, torque wrench, ruler, 5 kg push (a hand and a luggage scale work).

1. **HANG AN M6 EYE NUT ON EACH PROTRUDING AXLE END** (Ø20 eye over the Ø15 axle, threaded boss pointing rearward at the lug). Nothing to make — rev 004a replaced the tapped adjuster block with this catalog part.
2. **WELD THE LUGS** to each trailing fork-plate *outer face*: upright with an Ø6.6 clearance hole 11 mm above the plate, front face **34 mm behind the nominal axle centre** (behind the slot's rear end, forward of the plate tip — a tip-mounted lug puts the bolt head through the belt; Sheet 6 note). Weld all round.
3. **FIT THE IDLERS.** Leading wheel: Ø15 axle through the round bores + wheel + nylocks. Trailing wheel: axle through the slots, an eye nut on each protruding end, **draw bolts backed off**, axle at the most-forward slot position.
4. **FIT THE BELT FIRST** (§9.7), then come back here.
5. **TENSION:** thread each draw bolt through its lug into the block, then advance **both bolts evenly** — count the turns — until mid-span belt sag reads **10–15 mm under a 5 kg push**, arms at neutral. The bolt head pulls against the lug and draws the axle rearward.
6. **SQUARE THE AXLE:** equal turn counts both sides; measure axle-to-boss on both fork plates — equal within 0.5 mm, or the belt will walk.
7. **LOCK IT:** torque the axle nylocks, then run the jam nuts down against the lugs.
8. **RE-CHECK AFTER THE FIRST HOUR OF RUNNING** — a new belt seats and loosens (Sheet 6 note).

CHECK BEFORE MOVING ON

Sag 10–15 mm at mid-span, both draw bolts at the same turn count, axle square in the slot, nylocks torqued, jam nuts tight against the lugs, witness marks on.

COMMON MISTAKES

Advancing one draw bolt more than the other — the axle skews and the belt walks off within minutes (§10 bench spin catches this). Welding the lug at the plate tip instead of the 34 mm station — the bolt head then sits outside the belt's wrap radius and fouls the belt. Tensioning before the belt has been run around by hand. Skipping the 1-hour re-check on a new belt.

§9.7 FITTING THE BELT

The belt goes on last, with the tensioner fully slack. It stores real energy when levered — treat it with respect (§10 safety).

Tools: rubber or plastic tyre levers only. Eye protection. **Power off / battery out.**

1. **SLACK EVERYTHING:** trailing axle at its most-forward slot position, jack bolts backed off fully.
2. **ENGAGE THE SPROCKET FIRST.** Seat the belt lugs into the Ø180 10T sprocket at the top and check the guide lugs sit either side of the wheels' running faces.
3. **WALK THE BELT OVER THE LEADING IDLER,** rotating the assembly by hand so the lugs stay engaged on the sprocket.
4. **LEVER IT OVER THE TRAILING IDLER** with rubber/plastic levers — **never a screwdriver into the carcass**; a nicked cord line becomes a snapped belt under load.
5. **ROTATE TWO FULL PASSES BY HAND** and watch: lugs centred on the sprocket, belt centred on both idlers, no fold-over at the edges.
6. **TENSION** per §9.6 steps 4–7.

COMMON MISTAKES

Levering with screwdrivers (carcass damage you can't see). Fitting with any tension on the jack bolts. Working the belt anywhere near a powered motor — the sprocket-to-belt nip point pulls a glove in instantly (§10).

§9.8 PAINT, FINAL ASSEMBLY & TORQUE

Everything so far was trial-fitted. Now it comes apart one last time, gets painted, and goes together for real.

Tools: degreaser, masking tape, zinc-rich primer + topcoat, EP2 grease (+ gun if zerks fitted), torque wrench, paint pen.

1. **DISASSEMBLE COMPLETELY.** Shocks off, arms off, bushings out (press from the flange side), carriers off.
2. **MASK EVERY BORE AND SEAT:** pivot bores, bushing seats, axle-slot flats, wheel bores/slots, tab holes. Paint in a bore changes the fit you just built.
3. **DEGREASE, ZINC-RICH PRIMER, TOPCOAT** all steel (§8 finish spec). Let it cure fully.
4. **PRESS THE BUSHINGS BACK IN** (vise + socket) and re-verify the Ø16 axle turns by hand.
5. **FINAL-ASSEMBLE THE PIVOT STACK** in Sheet 2 order with EP2 grease on the axle, bushings and thrust washers. M16 nylock: snug to zero end-float, back ⅛ turn — arms must swing free.
6. **REFIT CARRIERS TO FORK, KEEL, SHOCKS, IDLERS, BELT, TENSION** — §9.4 → §9.5 → §9.6/§9.7 order.
7. **TORQUE EVERYTHING** (§8): M8 cl.10.9 — **30 N·m** · M10 — **60 N·m** · M12 — **105 N·m** · M16 pivot nylock — running fit as above, *never* torqued like a clamp bolt.
8. **WITNESS-MARK EVERY FASTENER** with the paint pen — one straight line across nut, bolt and parent metal. The §10 retorque schedule (1 h / 5 h / every 20 h) then takes a 2-minute glance.
9. **GREASE THE ZERKS** (if fitted) until fresh grease shows at the thrust washers.
10. **RUN THE §10 TEST PROTOCOL** — bench articulation, bench spin, loaded creep, ridden test — before the pod ever carries a rider.

COMMON MISTAKES

Painting the bushing seats or axle-slot flats (fits go loose or seize). Pressing bushings before paint bakes/cures. Torquing the M16 pivot — it's a running fit. Skipping witness marks: without them the retorque schedule is guesswork. Riding before §10.

§10 TEST PROTOCOL & SAFETY

1. **Bench articulation:** pod off the ground, cycle each arm full travel 50× by hand. No binding, no belt-lug contact with the sprocket at full bump (≥10 mm, Sheet 5), no shock coil-bind.
2. **Bench spin:** drive the pod unloaded at walking-pace RPM for 5 minutes. Watch belt tracking — it must not walk toward either edge. If it walks, the fork plates aren't parallel or the tension axle isn't square.
3. **Loaded creep:** vehicle weight on, no rider, walk-beside at <5 km/h over a 50 mm plank. Confirm articulation and belt retention.
4. **Ridden test:** flat ground, low speed, then progressively rougher terrain. Stop and inspect after 10 minutes: retorque everything, re-check belt tension (new belts loosen), look for cracked tack welds and shiny rub marks.
5. **Retorque schedule:** after 1 h, 5 h, then every 20 h. Witness-mark lines make this a 2-minute glance.

SAFETY

Tracked pods are finger-traps: never work near a powered, moving belt; the sprocket-to-belt nip point will pull a glove in instantly. Wear eye protection when levering the belt on — it stores real energy. And treat rev 001 as a prototype: first rides in an open area, helmet on, away from traffic and bystanders.

§11 GENERAL NOTES

- All dimensions mm unless noted. Formulas reference Sheet-0 datums; verify every derived number against the physical pod before cutting.
- Break all sharp edges 0.5 mm. Deburr every bore both sides.
- Fasteners class 10.9 (metric) / Grade 8 (US) minimum at pivot and shock points; all nuts nylock or thread-locked.
- This drawing set describes one pod. The carrier is a symmetric weldment — the same pieces serve both sides and both pods; and with both fork gaps re-measured at 140 (Rev 003) the two pods are fully identical — same pivot and keel lengths.
- Rev 001 supersedes the AI-concept sketch it was derived from. Known deltas: arms changed from single plates to fork weldments (keeps idlers centred on the belt), shock tab moved from 0.45·C to 0.68·C (sane spring rates), tensioner changed from eccentric cam to slot + jack bolt (easier to fabricate accurately), carrier bearing + anti-rotation link added (the concept had no torque path to the chassis).
- **Rev 001a** (from the 3D fit-check in the companion OpenSCAD model): shocks moved outboard of the carrier plates and the four cross-standoffs replaced by a single keel standoff below the pivot — both original placements collided with the sprocket's swept volume.
- **Rev 001b** (measured datums, 2026-07-12): track 1083×118 (60 mm pitch × 18 links) — idler spacing A now solved from belt length (≈ 243); sprocket OD 180; idlers Ø108×58 on 6302-2RS bearings → Ø15 axles. **Belt-clearance fix:** at full bump the belt bottom run rises ~32 mm and the 001a keel would have collided by ~40 mm — keel relocated between sprocket and pivot boss, pivot raised (P = B – 30), carrier tongue slimmed Ø56→Ø48. The OpenSCAD model prints a PASS/WARN clearance check per hanging part on every render.
- **Rev 001c**: pivot raised further (P = B – 38, near the 42 mm limit where the keel window closes) — lug-top margin at full bump grows 7.6→15.3 mm and the steel-touches-belt angle moves from ~+18.5° to ~+22°, buffering against shock-travel overshoot. Keel moved up 6 mm with it; keel-to-sprocket and keel-to-pivot-washer gaps added to the clearance guard. Cost accepted: belt-path variation through travel grows 3.6→5.7 mm, within tensioner range.
- **Rev 002** (owner interview): the pod is a **hub-motor scooter conversion** — motor inside the Ø180 10T sprocket on a static Ø10 flatted axle between fork legs (gaps: front 120, rear 140). Carriers redesigned as fork-hung torque-arm plates (identical symmetric part; axle slot + one M8 per leg); 6205 bearings, axle sleeve and anti-rotation link deleted; shocks moved inboard of the carriers (|z| = 73). Belt solver corrected to the cord line (10T × 60 = Ø191): A = 243 →

222.2, C = 117.4. Carriers now ride outside the belt width — the tongue-hits-track failure mode is geometrically impossible. New fit risk: 118 belt in the 120 front fork gap (1 mm/side).

- **Rev 002a** (2026-07-13, documentation only — no dimension changes): §7 expanded from a bare BOM into a full shopping guide (§7.1–§7.9: per-part specs, sources, search terms, substitutes, costs, receiving inspection); §9 expanded from the 10-step sequence into per-component build chapters (§9.1–§9.8: tools, steps, checks, common mistakes); print pagination CSS added so the PDF paginates one topic per page.
- **Rev 002b** (2026-07-13): Sheet-6 belt tensioner detailed and modeled explicitly in 3D — motorcycle chain-adjuster style: tapped adjuster block on each protruding axle end, M8 draw bolt through a lug welded to the fork-plate outer face (front face 34 mm behind nominal axle centre), head + jam nut on the lug's rear face; advancing the bolt *draws* the axle rearward. 3D fit-check moved the lug off the plate tip — a tip lug puts the bolt head outside the belt's 54 mm wrap radius at slack (through the belt). New .scad features: `tension_pos` parameter (0–25 slot take-up), `render_mode="tensioner"` labeled close-up, and a "draw-bolt head to belt wrap" PASS/WARN guard line. BOM #12 updated (M8×60 full-thread + jam nuts + 2 blocks).
- **Rev 002c** (2026-07-13, fork legs **measured: 4 mm** — the 30 mm placeholder is retired; only the guide-lug height remains open): carrier planes move in to |z| = 64 front / 74 rear (were 90/100). Pivot axle shortens — stack ≈162/182 → **M20×1.5 × 170 / 190** (were 220/240); fork bolts M8×30 (were ×45); keel tube 128/148 and rod ≈152/172 (were 180/205). **Shocks return outboard of the carriers** (|z| = 84/94, upper tabs on the carrier *outer* faces, lower sleeves ≈35): with the carriers at 64 there are only 5 mm between belt edge and plate — the rev-002 inboard placement physically no longer fits. Shock geometry in the side view (a, θ, eye position, motion ratio, spring rates) is unchanged. The .scad now picks the shock side automatically and echoes buy/cut lengths for axle, keel and sleeves per fork_gap.
- **Rev 003** (2026-07-20, owner changes): **pivot downsized M20 → M16** cl.10.9 (~9× margin over the full-bump load retained). Everything on the pivot cascades: bushings → SAE 841 flanged **16×22×20** (flange ≈Ø28×3), boss tubes → Ø32 OD × Ø22 ID (ream Ø22 H7), spacer tube → Ø22×3 wall, thrust washers → Ø30×Ø16×1.5, carrier pivot bores → Ø16 +0.1. Boss OD and the carrier window are unchanged; all belt-clearance guards still PASS (the spacer margin improves, Ø25 → Ø22). **Fork gaps re-measured: 140 mm front AND rear** (the earlier 120 front reading was wrong) — both pods are now identical: carriers |z| = 74, axle stack ≈182 → **M16 × 190** ×2, keel tube 148 + rod ≈172 ×2, shocks outboard at |z| = 94, lower sleeves ≈45, shock bolts ≈135. The 1 mm/side front belt-fit watch-item is retired: 11 mm per side, both pods.
- **Rev 003a** (2026-07-20, full 3D collision audit — the owner spotted the first two, the audit found the rest; every fix is now a PASS/WARN guard in the .scad): **(1) Pivot stack made**

continuous — the arm pack (~92 wide) never reached the carrier inner faces (148 apart): ~29 mm of bare axle per side meant the nylock had nothing to snug against and the arms could slide sideways. Bosses now point inboard (trailing) / outboard (leading), centre spacer ≈60 → ≈14, two new outboard sleeves ≈7, thrust washers 4 → 6. **(2) Shock bolt vs wheel** — the 0.68·C through-bolt passed 12 mm inside the Ø108 idler; moved to 0.485·C (4 mm clear). **(3) Shock coil vs carrier** — at 57° the Ø~44 spring overlapped the carrier tongue; θ → 68°, tab foot re-aimed at the hub boss, upper eye now (±52, +6); MR 0.57 → 0.45, springs ≈4.9×k_wheel (~620 lb/in in the worked example). **(4) Cross-brace vs wheel** — moved from a±25 to 28–58 (7 mm clear). **(5) Keel vs arm plates** — the keel at 104 ran through the Ø64 plate boss discs; discs slimmed to Ø44, keel tube Ø16 → Ø12×1.5 raised to 100 (4/4 mm, articulation-invariant). Ø30 shock-bolt lobe added to the arm profile. Open: guide-lug height still unmeasured — the diagonal-run lug clearance near full bump is the one check the guards do not yet cover.

- **Rev 004** (2026-07-21, flat-bar edition — owner request: every plate from off-the-shelf rectangular flat bar, straight cuts + drilled holes, no laser/waterjet): **arm plates → plain 40 × 6.35 flat bar** (trailing ≈185, leading ≈166). To fit the 40 mm bar the lower shock bolt moved onto the bar centreline (was y = +18), station a 0.485·C → **0.46·C = 54.0** (keeps 4 mm wheel clearance), θ 68° → **72°** (coil clears the carrier strip by 6 mm), cross-brace raised to y −18..−6, upper eyes at **(±53, −9)**; MR ≈ 0.44, springs ≈5.2 × k_wheel (~655 lb/in worked example — the owner's 100 kg/8.5 mm units still fit). **Carriers → 50 × 6 strip (≈225) + two 40 × 6 × 55 tab stubs** lap-welded on the OUTER face at hub level. Bonus: keel-to-arm gap grows 4 → 6 mm and the bar needs no tip rounding (half-width 20 stays inside the r34 lug wrap). All clearance guards re-run and PASS; steel order becomes ~1.8 m of flat bar, no plate stock. **Rev 004a** (same day): tensioner adjuster blocks replaced by **M6 lifting eye nuts** (Ø20 eye over the axle ends; owner-supplied) — draw bolts and jam nuts M8 → M6, lug holes Ø6.6; the last machined part in the tensioner is gone. **Rev 004b** (same day): shock eye bore measured — **Ø8, not the drawing default Ø10** — so the lower through-bolts, upper pins, and shock sleeves all became M8/ID9 (was M10/ID10) throughout §7.4/§7.7/§9.5; geometry (a, θ, MR) unchanged. **Rev 004c** (2026-07-22, real hardware confirmed): **pivot bolt** is the owner's actual part — M16 × 195, threaded 45 mm on ONE end only (a headed bolt, not a double-end stud). Smooth shank = 195−45 = 150, which clears the farthest bushing edge (145.85 from the head) by 4 mm — that margin sits inside the outboard-sleeve zone, not on a bearing surface, so it's safe; the nut fully engages 16 mm inside the threaded zone. Needs only **one M16 nylock per bolt** (earlier shopping-list drafts said 4 total, assuming a stud with a nut each end — corrected to 2, one per pod). **Boss tube** sourced at **31 OD × 22 ID** (was 32 OD spec) — wall 4.5 (was 5); arm-plate pivot holes 31.8 → **30.8**; weld shelf in the 40-wide bar actually grows slightly (4.0 → 4.6 mm/side). No clearance-guard impact — boss OD was never a guarded dimension.

APOLLO TRACK POD — ARTICULATED SPLIT-FRAME SUSPENSION · REV 004c · 2026-07-22 · COMPANION 3D MODEL: apollo_track_pod.scad (OpenSCAD) · ALL LOAD-BEARING DATUMS MEASURED — STILL OPEN: GUIDE-LUG HEIGHT. MEASURE IT, RE-RENDER THE GUARD, THEN CUT.